Isaiah Riddell

825-438-7165 | iriddell@student.ubc.ca | Linkedin | Github

Education

University of British Columbia Vancouver, BC

Bachelor of Applied Science, Engineering Physics | Cumulative Avg: 92.6% (4.33/4.33) *Sep. 2025 – Present*

Profile

UBC Engineering Physics student with a strong foundation in both software development and hardware integration. Proficient in Python, C, and MATLAB, offering hands-on experience in hardware prototyping, predictive modeling, and full-stack development. An adaptable problem-solver eager to collaborate with multidisciplinary teams and contribute impactful technical solutions from initial concept to deployment.

Skills

Programming Languages: Python, C, Java, HTML/CSS, JavaScript, Jinja2
Frameworks & Libraries: Flask, SQLAlchemy, Jinja2, jQuery, Bootstrap, Select2
Engineering & Data Software: MATLAB, Excel, R, SolidWorks
Hardware & Fabrication: ESP32 microcontrollers, Arduino, motor calibration, sheet metal fabrication, PID tuning

Technical Experience and Projects

Web Developer — Student Sign-In & Activity Reports Sep 2024 – Jun 2025

Renert School *Calgary, AB*

- Extended an existing Flask-based school management platform with a real-time student sign-in and activity report feature, working alongside another student developer and translating school administration's workflow needs into technical requirements
- Wrote the SQLAlchemy queries and filtering logic powering the report and activity-picker screens, enabling staff to instantly look up sign-in records by student, activity, and date range
- Built out the activity/class data model and Jinja2/Bootstrap templates (jQuery, Select2) for the reporting UI, contributing to a feature now in full, ongoing use by the after-school program

Custom ESP32-Based Quadcopter June 2026 – Aug. 2026

Independent Hardware Project

- Assembled and wired an ESP32 flight controller for a custom-built drone, resolving pin assignment conflicts to ensure proper system integration
- Calibrated motors and ran extensive bench failsafe checks prior to flight testing
- Tuned PID gains in Angle mode across multiple test flights, optimizing weight distribution and eliminating over-adjustment oscillation to achieve stable, responsive flight

IoT Sensor Predictive Modeling & ML Pipeline May 2026 – Aug. 2026

Independent Software Project

- Engineered an end-to-end classification pipeline in Python to predict industrial equipment failures from IoT sensor telemetry
- Executed rigorous data preparation including time-series feature extraction, correlation analysis, and one-hot encoding with strict train-test separation to prevent data leakage
- Implemented a Random Forest Classifier to leverage ensemble bagging and non-linear threshold splitting, bypassing feature scaling needs while maintaining robustness against correlated sensor signals
- Key Results: Achieved 100% test accuracy, using feature importance extraction to validate that predictions aligned with the underlying physical progression of electromechanical failures (temperature, current, and vibration)

Arduino-Controlled Mechanical Claw Jan. 2026 – Feb. 2026

Academic Engineering Project

- Designed and fabricated a functional mechanical claw for an engineering design competition, managing the project from initial concept to physical prototype
- Manufactured the structural components from sheet metal using manual hand tools, ensuring precision and mechanical reliability
- Integrated and programmed Arduino microcontrollers and electronic components to successfully actuate the claw for competition testing, validating the core electromechanical design